

MD. JEWEL

AI/ML Researcher || Software Engr. || U.S. SEC Filing Lead

✉ jewel15-8071@diu.edu.bd

🎓 [Google Scholar](#)

🐙 [GitHub](#)

🌐 [LinkedIn](#)

📁 [Portfolio](#)

👤 Research Profile

AI/ML researcher and software engineer with research experience in machine learning, deep learning, computer vision, medical AI, image segmentation, audio classification, predictive modeling, and applied AI. My published and ongoing work spans healthcare AI, computer vision, forecasting, AutoML, and intelligent decision-support systems. I also develop AI systems using Large Language Models (LLMs), Transformers, Retrieval-Augmented Generation (RAG), semantic retrieval, and generative AI, with hands-on experience in Python, Java, PyTorch, TensorFlow, and AI-powered automation.

Research Interests: Machine Learning • Deep Learning • LLMs • Transformers • RAG • Generative AI • Multimodal AI • Computer Vision • Medical AI • Image Segmentation • Foundation Models • Audio Intelligence.

📁 Research Experience

🌟 Applied Research in Software & IT (CARSIT) (2022 - Current)

Responsibilities (Volunteer):

- Conduct research in machine learning, deep learning, computer vision, data science, and applied AI.
- Lead research tasks including dataset preparation, model development, experimentation, analysis, and research paper writing.
- Coordinate co-authors and often handle manuscript submission, journal communication, revisions, and corresponding-author responsibilities.

🌟 DIU NLP Research Lab (2019 - 2020)

Responsibilities: Primary responsibility involved active participation in research and publications, resulting in the successful publication of a few research papers.

⚙️ Education

🌟 B.Sc. in Computer Science & Engineering | CGPA: 3.93 / 4.00 (2016 - 2020)

Daffodil International University

🌟 **Undergrad Thesis:** Instrumental Sound Classification; A way to speech recognition using Convolutional & Recurrent Neural Network.

📁 Work Experience

🌟 U.S. SEC Filing Lead for Barclays Bank. PLC (2023 – Current)

at Compliance Services Ltd. (WSD, remote)

Responsibilities:

- Lead and participate in U.S. SEC EDGAR filing, review, and quality-assurance activities supporting Barclays Bank PLC, and U.S. Bancorp.
- Develop AI-powered tools and automation solutions using Java and Python to streamline filing, conversion, review, and repetitive operational workflows.
- Apply software engineering and AI-assisted approaches to improve process efficiency and filing accuracy.
- Train and mentor new team members on SEC filing procedures, internal systems, and operational workflows.

-  **U.S SEC Filer**
 at **Apurba Technologies Ltd.** (2020 - 2022)
Responsibilities: Handling the conversion of docx files to HTML code, making necessary modifications, and overseeing filing and reviewing processes.
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-  **Student Prefect**
 at **Daffodil International University** (2018 - 2019)
Responsibilities: Collaborating with course teachers to maximize the throughput of the course by providing necessary guidelines, materials, and assistance to the students.

Teaching Experience

-  **Lecturer at various coaching centers** (2015 - Current)
Responsibilities: Instructing HSC level students at private coaching center in Networking, C programming, HTML, and Database Management Systems (MySQL), Digital Circuit Design among other subjects.

Selected Publications

-  **Md. Jewel, Karim, Kazy, & Shabiul (2026).** “A Hybrid Framework for Mpox Diagnosis and Severity Assessment via Symptom Classification and Skin Lesion Segmentation.” *PeerJ Computer Science*. **Accepted for publication.**
-  **Rezaul, Jewel, and et al.(2025),** ‘A Comparative Study of Predictive Analysis Using Machine Learning Techniques: Performance Evaluation of Manual and AutoML Algorithms.’ *International Journal of Advanced Computer Science & Applications*, 16(1), DOI: [10.14569/IJACSA.2025.0160102](https://doi.org/10.14569/IJACSA.2025.0160102)
-  **Karim, Jewel, and et al(2024)** ‘Enhancing audio classification through MFCC feature extraction and data augmentation with CNN and RNN models’. *International Journal of Advanced Computer Science and Applications*, 15(7). DOI: [10.14569/IJACSA.2024.0150704](https://doi.org/10.14569/IJACSA.2024.0150704)
-  **M. Jewel, M. I. Hossain and T. H. Tonni,** "Bengali Ethnicity Recognition and Gender Classification using CNN & Transfer Learning," *2019 8th International Conference System Modeling and Advancement in Research Trends (SMART)*, Moradabad, India, 2019, pp. 390-396, (IEEE) DOI: [10.1109/SMART46866.2019.9117549](https://doi.org/10.1109/SMART46866.2019.9117549)
-  **M. Jewel, Al Foysal,** “[Instrument Sound Classification](#)”, published at DSpace Repository, Daffodil International University, 2020.

Manuscripts Under Peer Review

-  **Md. Jewel, Khan, & Rezaul (2026).** “Moving Beyond Student Feedback: A Seven-Factor, Fairness-Aware, and Tamper-Evident Framework for Faculty Evaluation.” **Under peer review.**
-  **Abid, Karim, Md. Jewel, Siddiquee, & Shabiul (2026).** “Multivariate Time Series Modeling of Ethereum Price Dynamics Incorporating Market Indicators.” **Under peer review.**
-  **Sohan, Md. Jewel, & Rezaul (2026).** “Does Model Complexity Improve Consumer Price Forecasting? Evidence from Data-Scarce, Crisis-Prone Economies.” **Under peer review.**
-  “FORGE-Vision: A Hybrid Computer Vision and Generative AI System for Real-Time and Simulation-Based Employee Training.” **Manuscript prepared for submission.**



TECHNICAL & RESEARCH SKILLS

- ✚ **AI & Machine Learning:** Machine Learning, Deep Learning, Computer Vision, Image Segmentation, Multimodal AI, Medical/Healthcare AI, Audio Classification, Time-Series Forecasting, AutoML, Generative AI
- ✚ **LLMs & Intelligent Systems:** Large Language Models (LLMs), Transformers, Retrieval-Augmented Generation (RAG), Semantic Search/Retrieval, Vector Embeddings, Local LLM Inference, AI Agents/Tool Integration
- ✚ **Research Methods:** Dataset Preparation & Curation, Model Training & Fine-Tuning, Transfer Learning, Experimental Design, Model Evaluation, Comparative Analysis, Research Prototyping
- ✚ **Programming:** Python, Java, C/C++, JavaScript/TypeScript, SQL
- ✚ **AI/ML Frameworks:** PyTorch, TensorFlow, Keras, Hugging Face Transformers, FAISS
- ✚ **Software & Backend:** FastAPI, Flask, Django, Spring Boot, Node.js, NestJS, React, Next.js
- ✚ **Data & Development Tools:** PostgreSQL, MySQL, SQLite, Git/GitHub, Docker, Linux/Bash, Selenium



SELECTED AI & SOFTWARE PROJECTS

- ✚ **NexusForge AI:** Fully local generative AI system for image and video generation/editing, supporting text-to-image, image-to-image, text-to-video, image-to-video, and local audio generation. **Technologies:** *Stable Diffusion, AnimateDiff, Vulkan, Kokoro TTS, FFmpeg*
- ✚ **NexusMind AI:** Fully local AI assistant/workstation with local LLM inference, Retrieval-Augmented Generation (RAG), semantic document search, code assistance, workspace/file analysis, and tool execution. **Technologies:** *LLMs, Transformers, RAG, llama.cpp, Qwen, FAISS, FastAPI, Next.js, SQLite, Vulkan*
- ✚ **File Reviewer:** Document-review tool for validating converted HTML against source documents, reducing reliance on manual inspection. **Technologies:** *JavaFX*
- ✚ **Junk Reducer:** Automation tool for identifying and removing unnecessary HTML code and assisting with correct linking before SEC EDGAR filing. **Technologies:** *Python, Regular Expressions, Flask*
- ✚ **Conversion Tool:** Configurable document-conversion system designed to transform Word documents into SEC-compatible HTML while accommodating evolving EDGAR requirements. **Technologies:** *JavaFX, Maven*
- ✚ **Info Tracker:** Filing-information management system for tracking submission timing, tracked-change information, filer details, SEC links, and operational metadata. **Technologies:** *Spring Boot, React, MySQL*
- ✚ **Smart Aud:** Audio-analysis application for speech/audio recognition using voice or audio input. **Technologies:** *Python, Flask, HTML/CSS, JavaScript*
- ✚ **DIU AI Hub:** AI-focused platform enabling students to upload AI-related projects and publish technical ideas and posts. **Technologies:** *Python, Flask, MySQL, JavaScript*



Honors & Awards

- ❖ **Best Student Award** among Inter Cantonment Schools (Dhaka)
- ❖ **BattleBot Championship**, Daffodil International University
- ❖ **Tuition Fee Waiver (75%)**, Daffodil International University